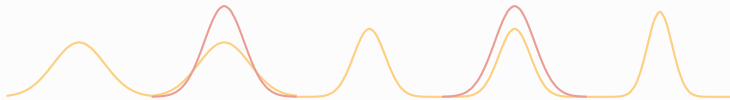


Gated Data-Space Inversion for Reservoir Prediction

Reducing variance loss in posterior

Masters dissertation Thiago Tomás de Paula



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Graduate Program in Mechanical Engineering

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Introduction



The master's program in Mechanical Engineering at University of Brasília requires

- the completion of a minimum of 360 hours of coursework (24 credits)
- at least one publication in a national or international conference/journal
- **presentation and approval of the master's dissertation**



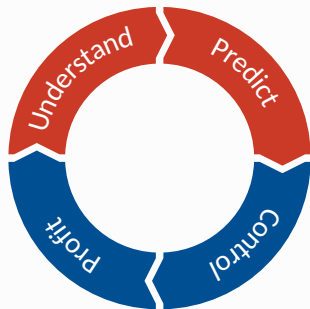
... plus 36 credits. Next up: two research projects (ongoing since December 2025).



As subsurface geological formations, hydrocarbon reservoirs are extremely difficult systems to handle: they are nonlinear, stochastic, multivariate, and difficult to sensor.

However, they are also extremely profitable.

The better you ***understand*** a reservoir, the better you ***predict*** its production/behavior, the more reliable your ***control***, and the greater your ***revenue***.



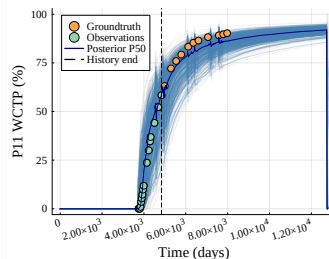
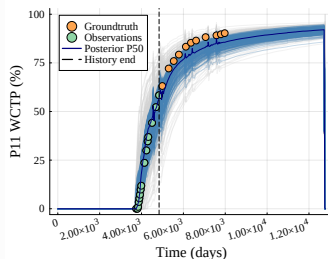
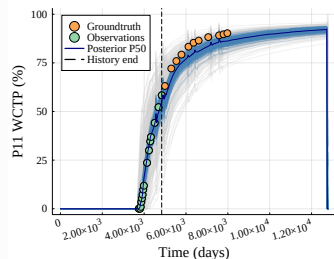




Data-Space Inversion (DSI) [Sun:2017] updates the predictions of reservoir model realizations directly, doing away with sluggish model updates.

Ensemble Smoother with Multiple Data Assimilation (ESMDA) [Emerick:2013], a seminal Model-Space algorithm, has been adapted to DSI, resulting in **DSI-ESMDA** [Lima:2020].

Despite recent advances [Jiang:2021, Xiao:2023, Jiang:2024], DSI-ESMDA remains susceptible to excessive variance loss. This work proposes **Gated DSI-ESMDA**, an extension of DSI-ESMDA with a gating mechanism for variance recovery.





Theory framework



Prediction forward model

The prediction forward model, or forward model, is the underlying mapping between a model's parameters, controls and initial states to its observable features.

Denoting by $\mathbf{m}$ the vector of model parameters, and by $\mathbf{u}$ the set of controls applied on the field, and by $\mathbf{d}$ the reservoir production within the time frame of the controls, one can write

$$\mathbf{d} = \mathbf{g}(\mathbf{m}, \mathbf{u}; \mathbf{x}_0) \quad (1)$$

with $\mathbf{x}_0$ the initial value of reservoir states (usually omitted).



Data features

In this work, the features inside **d** are the following well measurements:

- | | | |
|----------------------------------|---------------|-----------------------|
| ▷ oil and water production rates | (OPR and WPR) | [m ³ /day] |
| ▷ water injection rate | (WIR) | [m ³ /day] |
| ▷ bottomhole pressure | (BHP) | [Bar] |
| ▷ water cut percentage | (WCTP) | [%] |

Field oil production total

Field oil production total (FOPT, SI unit of m³) is also relevant to this work, despite not being part of **d**.



The key assumption behind the forward model is that it is deterministic. Implementation wise, $\mathbf{g}$ consists of evaluating a set of equations at each gridblock of the model.

Each evaluation evolves the model a little further in time, until the scheduled time period is appreciated, concluding the process.

Black-oil equations are the most common implementations of the forward model [Rasmussen:2021] due to their fair simplicity and accuracy. The fluid is assumed composed of three liquid phases, water, oil and gas.



A given phase α 's component mass A_α satisfies the conservation equation

$$\frac{\partial}{\partial t} (\phi_{\text{ref}} A_\alpha) + \nabla \cdot \mathbf{u}_\alpha + q_\alpha = 0, \quad (2)$$

where

$$A_w = m_\phi b_w s_w, \quad \mathbf{u}_w = b_w \mathbf{v}_w, \quad (3a)$$

$$A_o = m_\phi (b_o s_o + r_{og} b_g s_g), \quad \mathbf{u}_o = b_o \mathbf{v}_o + r_{og} b_g \mathbf{v}_g, \quad (3b)$$

$$A_g = m_\phi (b_g s_g + r_{go} b_o s_o), \quad \mathbf{u}_g = b_g \mathbf{v}_g + r_{go} b_o \mathbf{v}_o. \quad (3c)$$

and the phase fluxes are given by Darcy's law:

$$\mathbf{v}_\alpha = -\lambda_\alpha \mathbf{K} (\nabla p_\alpha - \rho_\alpha \mathbf{g}). \quad (4)$$

Finally, the following closure equations are imposed:

$$s_w + s_o + s_g = 1, \quad p_{c,ow} = p_o - p_w, \quad p_{c,og} = p_o - p_g. \quad (5)$$



Let $\mathbf{d}$ denote the true production data vector of a reference reservoir across time, free of observation error. Data is separated into a history period and a validation period, so that

$$\mathbf{d} = \begin{bmatrix} \mathbf{d}_h \\ \mathbf{d}_v \end{bmatrix} \quad (6)$$

with $\mathbf{d}_h$ the ground-truth history data, and $\mathbf{d}_v$ the ground-truth validation data.

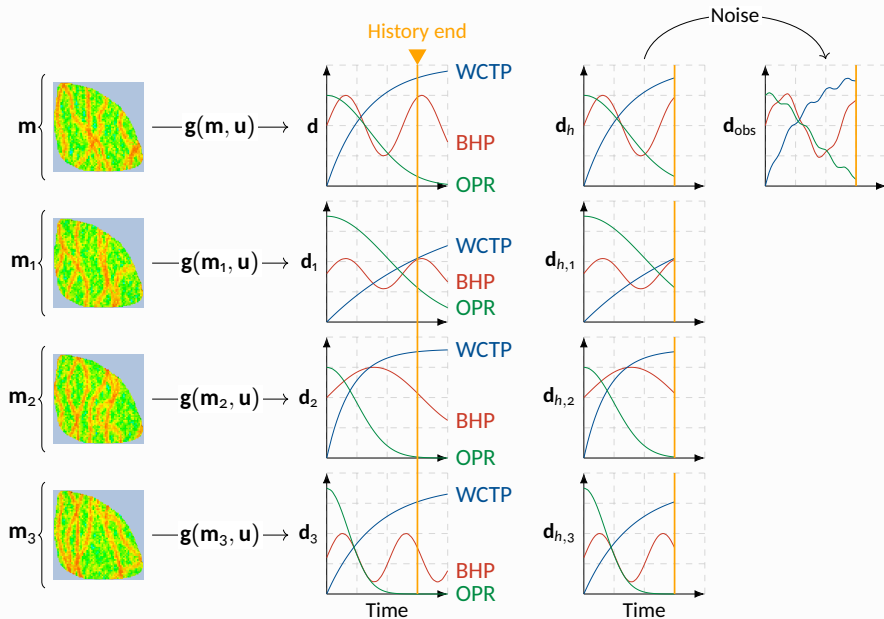
Ground-truth data come from the forward model,

$$\mathbf{d} = \mathbf{g}(\mathbf{m}, \mathbf{u}). \quad (7)$$

and observed data, from a corruption of $\mathbf{d}_h$:

$$\mathbf{d}_{\text{obs}} = \mathbf{d}_h + \mathcal{N}(\mathbf{o}, \mathbf{C}_e), \quad (8)$$

where $\mathbf{C}_e$ is the noise covariance matrix.





DSI-ESMDA can be conceptualized as a Bayesian approach to the forecasting problem, aiming to estimate the underlying conditional probability distribution of predictions given the reference data.

The DSI-ESMDA problem

Find the posterior distribution

$$p(\mathbf{d}^* \mid \mathbf{d}_{\text{obs}}) \propto \mathcal{L}(\mathbf{d}_{\text{obs}} \mid \mathbf{d}^*)p(\mathbf{d}^*). \quad (9)$$

where $\mathbf{d}^*$ maximizes $p(\mathbf{d}^* \mid \mathbf{d}_{\text{obs}})$.

The algorithm returns samples from this posterior distribution, outputting a range of values rather than a single value. While this makes evaluating the algorithm less straightforward, it allows for quantification of uncertainty.



Ensemble Smoothers (ES) share this approach at a conceptual level, but they perform a single maximization. This leads to poor matches to real data if assumptions on the likelihood $\mathcal{L}$ are not met.

Multiple Data Assimilation (MDA) mitigates this problem by making N_a optimizations with resampled reference data:

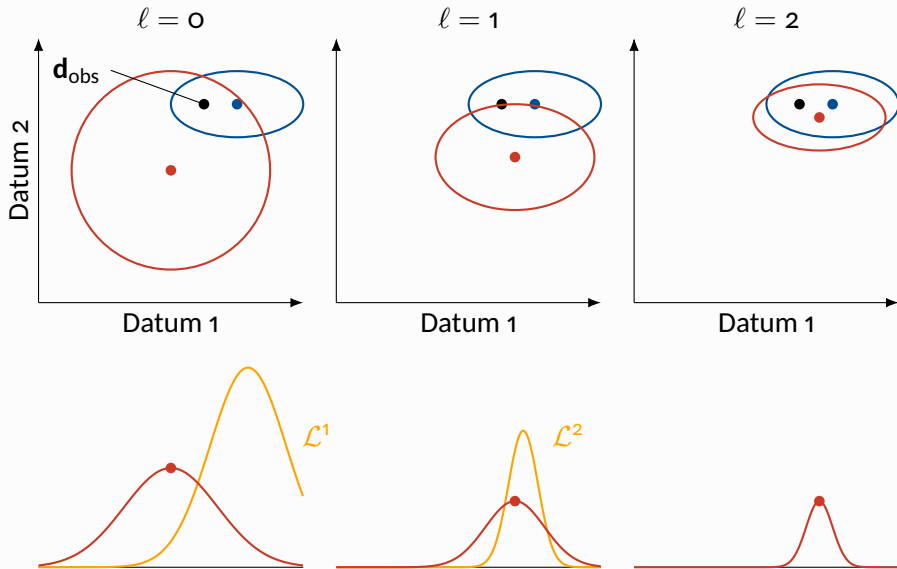
$$p^1(\mathbf{d}^* \mid \mathbf{d}_{\text{obs}}^1) \propto \mathcal{L}(\mathbf{d}_{\text{obs}}^1 \mid \mathbf{d}^*)p(\mathbf{d}^*) \quad (10)$$

$$p^2(\mathbf{d}^* \mid \mathbf{d}_{\text{obs}}^2, \mathbf{d}_{\text{obs}}^1) \propto \mathcal{L}(\mathbf{d}_{\text{obs}}^2 \mid \mathbf{d}^*)p^1(\mathbf{d}^* \mid \mathbf{d}_{\text{obs}}^1) \quad (11)$$

$$\vdots$$

$$(12)$$

$$p^{N_a}(\mathbf{d}^* \mid \mathbf{d}_{\text{obs}}^{N_a}, \dots, \mathbf{d}_{\text{obs}}^1) \propto \mathcal{L}(\mathbf{d}_{\text{obs}}^{N_a} \mid \mathbf{d}^*)p^{N_a-1}(\mathbf{d}^* \mid \mathbf{d}_{\text{obs}}^{N_a-1}, \dots, \mathbf{d}_{\text{obs}}^1) \quad (13)$$





MDA makes the following suppositions on the likelihoods:

- ▷ $\mathcal{L}^\ell$ is Gaussian
- ▷ $\mathcal{L}^\ell$ has covariance $\mathbf{C}_e^\ell = \alpha^\ell \mathbf{C}_e$, $\alpha^\ell > 0$

This leads to

$$\mathcal{L}^\ell(\mathbf{d}_{\text{obs}}^\ell \mid \mathbf{d}^*) \propto \exp \left(-\frac{1}{2} (\mathbf{d}_{\text{obs}} - \mathbf{H}\mathbf{d}^*)^\top \frac{\mathbf{C}_e^{-1}}{\alpha^\ell} (\mathbf{d}_{\text{obs}} - \mathbf{H}\mathbf{d}^*) \right), \quad (14)$$

$\mathbf{H}$ a matrix that extracts $\mathbf{d}_h^*$ from $\mathbf{d}^*$.

Equivalence to ES

For Gaussian likelihoods, MDA is equivalent to ES if

$$\sum_{\ell=1}^{N_a} \frac{1}{\alpha^\ell} = 1, \quad (15)$$

highlighting the importance of having inflation factors on the covariance.



The priors are also supposed Gaussian:

$$p^\ell(\mathbf{d}^* \mid \mathbf{d}_{\text{prior}}^\ell) \propto \exp \left(-\frac{1}{2} (\mathbf{d}^* - \mathbf{d}_{\text{prior}})^\top (\mathbf{C}_p^\ell)^{-1} (\mathbf{d}^* - \mathbf{d}_{\text{prior}}) \right), \quad (16)$$

where

- ▷ $\mathbf{d}_{\text{prior}}^\ell$ is the prediction with highest belief of occurring prior to assimilating $\mathbf{d}_{\text{obs}}^\ell$
- ▷ $\mathbf{C}_p^\ell$ is the prior covariance matrix (i.e., $\mathbf{d}^* \sim \mathcal{N}(\mathbf{d}_{\text{prior}}^\ell, \mathbf{C}_p^\ell)$)



The maximum *a posteriori* (MAP) estimate of the ℓ -th assimilation becomes

$$\mathbf{d}_{\text{map}}^{\ell+1} = \mathbf{d}_{\text{map}}^{\ell} + \mathbf{K}^{\ell}(\mathbf{d}_{\text{obs}}^{\ell} - \mathbf{d}_{h,\text{map}}^{\ell}), \quad (17a)$$

$$\mathbf{K}^{\ell} = \mathbf{C}_p^{\ell} \mathbf{H}^{\top} (\mathbf{H} \mathbf{C}_p^{\ell} \mathbf{H}^{\top} + \alpha^{\ell} \mathbf{C}_e)^{-1} \quad (17b)$$

with $\mathbf{d}_{\text{map}}^0 = \mathbf{d}_{\text{prior}}^0$.

Using an ensemble of N_e model realizations, one can estimate the prior covariances with $\Delta \mathbf{D} = [\mathbf{d}_1 - \bar{\mathbf{d}} \quad \dots \quad \mathbf{d}_{N_e} - \bar{\mathbf{d}}] / \sqrt{N_e - 1}$:

$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^{\ell} + \mathbf{K}^{\ell}(\mathbf{d}_{\text{obs}}^{\ell} + \sqrt{\alpha^{\ell}} \mathbf{e}_j^{\ell} - \mathbf{d}_{h,j}^{\ell}), \quad (18a)$$

$$\mathbf{K}^{\ell} = \Delta \mathbf{D}^{\ell} \Delta \mathbf{D}_h^{\ell} (\Delta \mathbf{D}_h^{\ell} \Delta \mathbf{D}_h^{\ell} + \alpha^{\ell} \mathbf{C}_e)^{-1}. \quad (18b)$$

$\mathbf{d}_{\text{obs}}^{\ell} + \sqrt{\alpha^{\ell}} \mathbf{e}_j^{\ell}$, with $\mathbf{e}_j^{\ell} \sim \mathcal{N}(\mathbf{0}, \mathbf{C}_e)$, is a way of sampling $\mathbf{d}_{\text{obs}}^{\ell} \sim \mathcal{N}(\mathbf{d}_{\text{obs}}, \alpha^{\ell} \mathbf{C}_e)$.



Covariance localization, or simply localization, is an *ad hoc* procedure employed to mitigate the effects of small ensembles [Emerick:2025].

The influence a datum has on another datum is weighted in an inverse manner to the distance between the points.

Localization is applied to the Kalman-like gain, so that the base equation becomes:

$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^{\ell} + (\mathbf{R} \circ \mathbf{K}^{\ell})(\mathbf{d}_{\text{obs}} + \sqrt{\alpha^{\ell}} \mathbf{e}_j^{\ell} - \mathbf{d}_{h,j}^{\ell}), \quad (19)$$

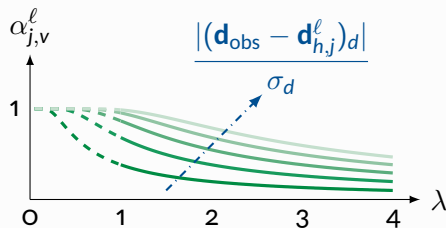
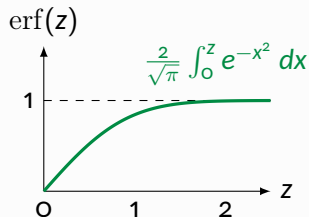


This work introduces gating α on the innovation vector:

$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^{\ell} + (\mathbf{R} \circ \mathbf{K}^{\ell})[\alpha_j^{\ell} \circ (\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,j}^{\ell})], \quad (20a)$$

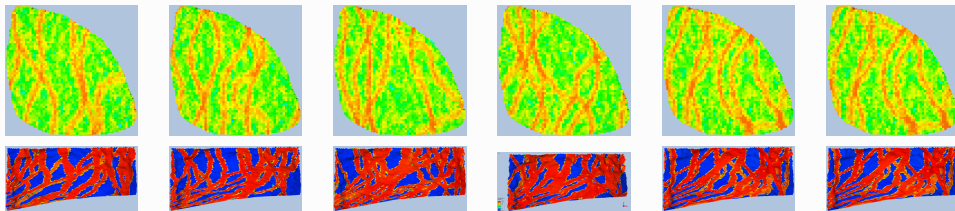
$$\alpha_{j,d}^{\ell} = \text{erf} \left(\frac{1}{\lambda \sqrt{2}} \frac{|(\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,j}^{\ell})_d|}{\sigma_d} \right). \quad (20b)$$

The gating recovers variance loss in the posterior ensemble by diminishing Kalman-like updates on reasonably matched data. The smaller the mismatch of a measurement relative to the noise at that datum, the smaller the update.





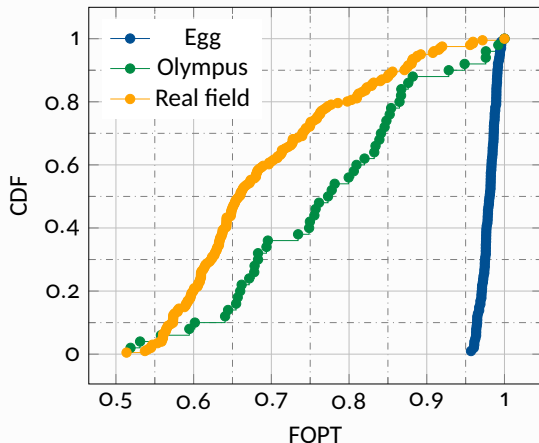
Methods

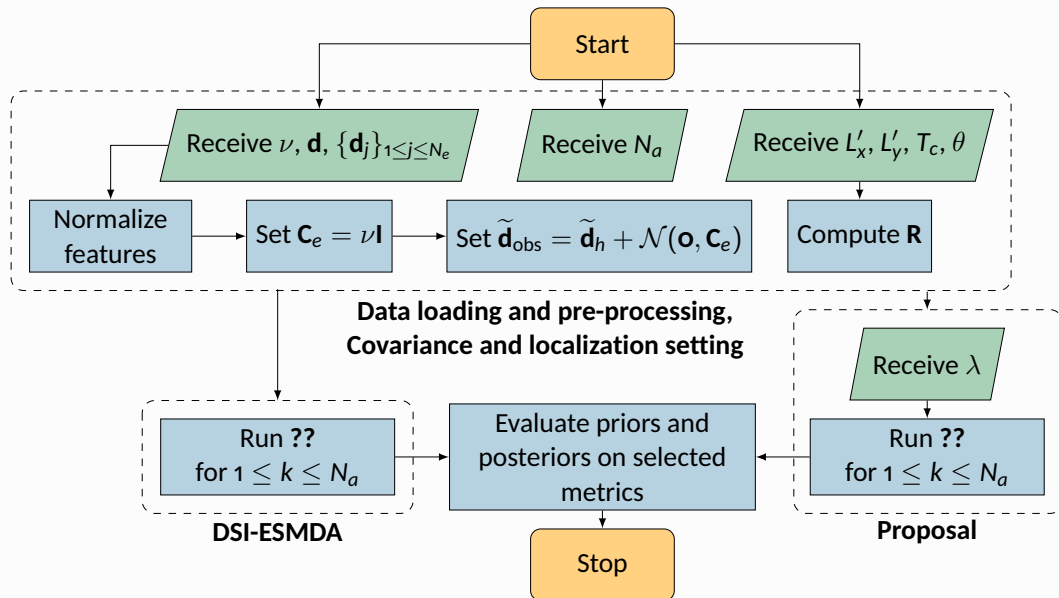


Property	Egg	Olympus	Real field	Unit
Rock compressibility	0.0	1.42×10^{-10}	-	1/Pa
Water compressibility	1.0×10^{-10}	3.97×10^{-10}	-	1/Pa
Oil dynamic viscosity	5.0×10^{-3}	$2.8-3.5 \times 10^{-3}$	-	Pa·s
Water dynamic viscosity	1.0×10^{-3}	0.398×10^{-3}	-	Pa·s
Well diameter	20	19	-	cm
Wells	8(P) 4(I)	11(P) 7(I)	16(P) 8(I)	-
Active cells	18,533	192,750	-	-
Cell size	$8 \times 8 \times 4$	$50 \times 50 \times 3$	-	m×m×m
Ensemble size	101	50	201	-



The field oil production total (FOPT) is a measurement commonly used to compare ensembles since ground-truth and realizations are each represented by a single value: how much oil was produced within the period of interest.







Four metrics are considered on the evaluation of posterior/prior distributions [Emerick:2025]: history observation coverage, validation observation coverage, mean squared error, and data-mismatch.

Denoting an ensemble by $\mathbf{Y} = [\mathbf{y}_1 \ \cdots \ \mathbf{y}_{N_e}]$, these metrics are defined as:

$$\text{OC}_h(\mathbf{Y}) = \frac{1}{N_h} \sum_{i=1}^{N_h} \mathbb{1} \left(\min(\mathbf{Y})_i \leq d_{h,i} \leq \max(\mathbf{Y})_i \right) \quad (21)$$

$$\text{OC}_v(\mathbf{Y}) = \frac{1}{N_v} \sum_{i=1}^{N_v} \mathbb{1} \left(\min(\mathbf{Y})_i \leq d_{v,i} \leq \max(\mathbf{Y})_i \right) \quad (22)$$

$$\text{MSE}(\mathbf{Y}) = \frac{1}{N_h} (\mathbf{d}_h - \bar{\mathbf{y}})^\top \mathbf{C}^{-1} (\mathbf{d}_h - \bar{\mathbf{y}}), \quad \mathbf{C} = \Delta \mathbf{Y} \Delta \mathbf{Y}^\top + \mathbf{C}_e \quad (23)$$

$$\mathcal{O}(\mathbf{y}_j) = \frac{1}{2N_h} (\mathbf{d}_h - \mathbf{y}_{j,h})^\top \mathbf{C}_e^{-1} (\mathbf{d}_h - \mathbf{y}_{j,h}) \quad (24)$$



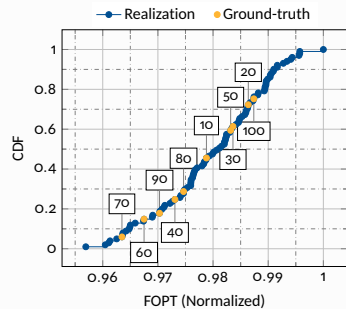
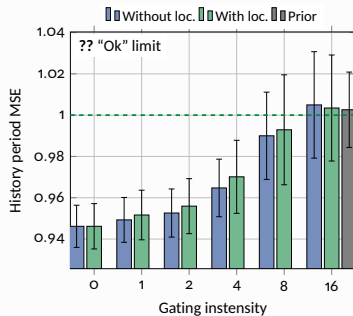
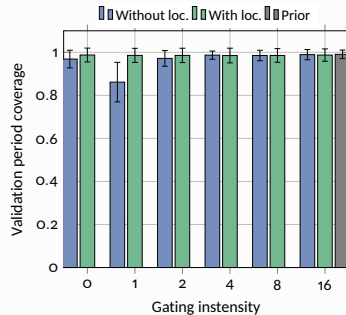
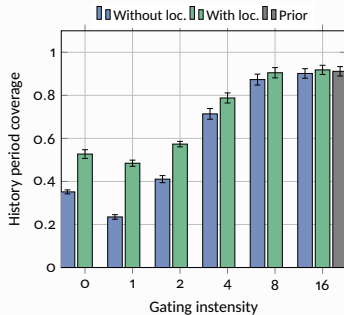
Reservoir	N_e	Number of time steps	Time step size	Simulation life	History split
Egg	100	120	30 days	3600 days (9.8 years)	60%
Olympus	49	20	1 calendar year	7305 days (20 years)	60%
Real field	200	510	1 calendar month	12785 days (35 years)	60%*

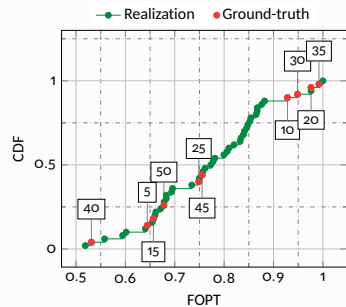
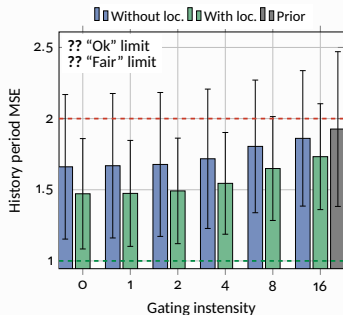
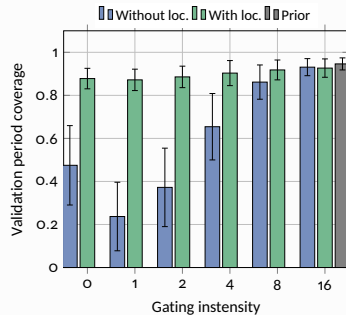
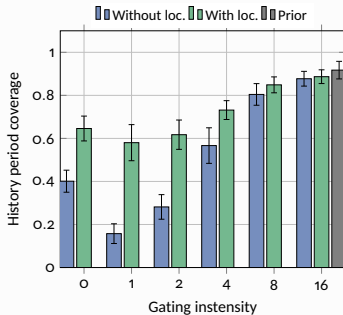
Reservoir	Critical length in x	Critical length in y	Critical time	Anisotropy
Egg	100	100	3000 days	0°
Olympus	100	100	7200 days	0°
Real field	-	-	10500 days	0°

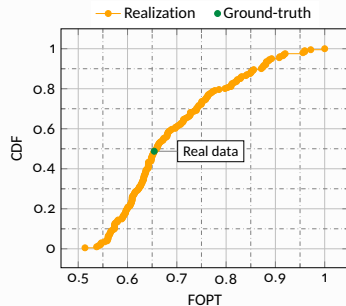
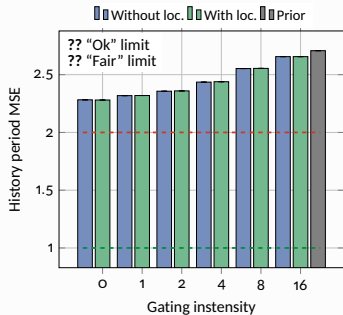
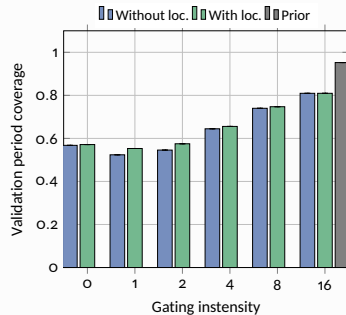
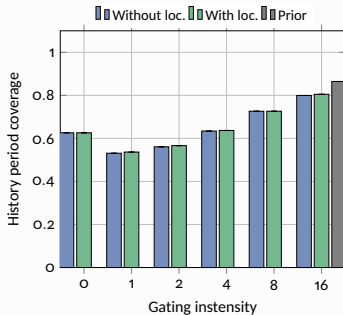
Reservoir	Well features used Producer	Well features used Injector	Number of wells Producer	Number of wells Injector	Total features
Egg	OPR, WPR	BHP	4	8	16
Olympus	OPR, WPR	WIR	11	7	29
Real field	BHP, WCTP	BHP	16	8	39 (a WCTP was constant)



Results









OC, MSE and mismatch increase

Increase in gating intensity tended to increase observation coverage, mean squared error, and data mismatch, albeit with case-dependent sensitivities.

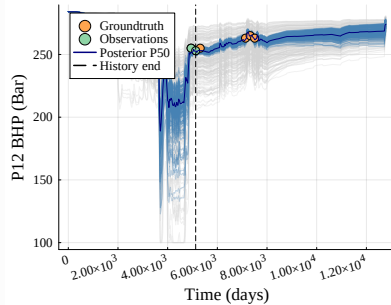
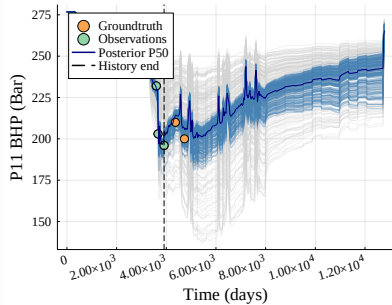
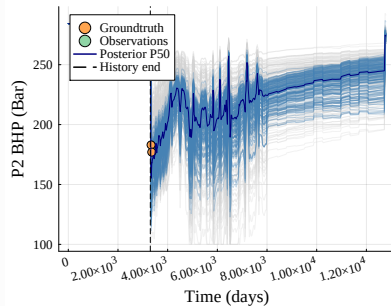
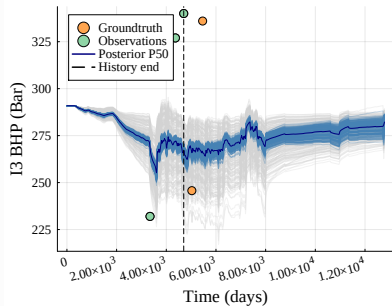
- ▷ Gating reduces posterior variance loss, so increase in coverages, but also MSE and mismatch, are expected as gating intensifies.
- ▷ Egg standard deviations are lesser than Olympus on two accounts: reservoir complexity, and ground-truth representativeness. Olympus' ground-truths were somewhat biased.



Mixed effects of localization

Localization yielded mixed results: for the Egg and real field cases, effects on coverage and MSE were minimal, especially for intense enough gatings, while the effect in the Olympus case was drastic.

- ▷ **Egg case:** narrow FOPT range and roughly uniform sampling of ground-truths leave no ground-truth unrepresented, reduced dimensionality help with covariance estimation.
- ▷ **Olympus case:** much more complex while also having a much smaller ensemble and a lot more features, aggravating covariance estimation. Localization was conceived exactly for such situations, so it is not surprising it was effective here.
- ▷ **Real field case:** peak complexity and dimensionality, but also ensemble size. Localization may have inadequately adjusted, and some features are not well-represented.





Mild and small gating intensities versus standard DSI-ESMDA

Mild gating intensities (2 in the Egg and Olympus cases, 4 in the real field case) produced similar results to the standard DSI-ESMDA for same localization configuration, while small gating (1 for the Egg case, 1 and 2 for the Olympus and real field cases) had worse coverages than the standard DSI-ESMDA.

- ▷ Gating intensity “reverts” the posterior to the prior. If the posterior given by the standard DSI-ESMDA is somewhere in that continuum, then it stands to reason that Gated DSI-ESMDA could approximate it simply by tuning the gating intensity.
- ▷ Since Gated DSI-ESMDA does away with observation data resampling, small gatings possibly underestimate uncertainty (i.e., likelihood covariance) at least when compared to standard DSI-ESMDA, producing posteriors that are not well-matched to data.



- ▷ This work introduced Gated DSI-ESMDA, an extension of DSI-ESMDA with a nonlinear innovation gating mechanism that increases posterior variance the more intense the gating.
- ▷ Observation coverages of the posterior increased with the gating intensity for all case studies, albeit under different sensitivities.
- ▷ Small gatings tend to underestimate uncertainty, while mild gatings produced similar results to standard DSI-ESMDA.
- ▷ The gating mechanism in of itself can be easily adapted into model-space methods such as ES-MDA.
- ▷ Gated DSI-ESMDA remains susceptible to poor feature selection and localization tuning, as well as insufficient ground-truth representativeness, high data dimensionality, and reduced ensemble size. Future research should address these problems.

Bibliography

Gated DSI-ESMDA

Reducing variance in variational posteriors

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“It’s hard to make predictions, specially about the future”

— Yogi Berra